

De Moivre's theorem

Key Points

De Moivre's theorem states

$$(\cos x + i \sin x)^n = \cos nx + i \sin nx$$

This allows us to get

$$\cos nx = \frac{z^n + z^{-n}}{2}$$

$$\sin nx = \frac{z^n - z^{-n}}{2i}$$

Where $z = \cos x + i \sin x$

Which can be used to express a power trigonometry as multiple angles

Basic Skills

Q1. Use de Moivre's theorem to simplify the following

a) $(\cos x + i \sin x)^3$

b) $\cos 6x + i \sin 6x$

c) $(\cos x + i \sin x)^{-5}$

Q2. Use de Moivre's theorem to simplify $\cos 5x - i \sin 5x$

Q3. Simplify the following to $a + bi$

a) $\left(\cos \frac{2}{3}\pi + i \sin \frac{2}{3}\pi\right)^3$

b) $\left(2 \cos \frac{1}{4}\pi + 2i \sin \frac{1}{4}\pi\right)^4$

c) $(2\cos 50^\circ + 2i \sin 50^\circ)^{-5}$

Competency Skills

Q4. a) Show that $\cos^3 x = \frac{1}{8}(z + z^{-1})^3$

b) Hence show that $\cos^3 x = \frac{1}{8}(z^3 + 3z + 3z^{-1} + z^{-3})$

c) Finally show that $\cos^3 x = \frac{1}{4}\cos 3x + \frac{3}{4}\cos x$

Q5. Show that $\sin^5 x = \frac{1}{16}\sin 5x - \frac{5}{16}\sin 3x + \frac{5}{8}\sin x$

Q6. Prove de Moivre's theorem using Euler's identity, that is use $e^{ix} = \cos x + i \sin x$

Q7. Simplify the following to $a + bi$

a) $(i - 2)^5$

b) $(\sqrt{3} + i)^{12}$

Advanced Skills

Q8. a) Show that $\cos^6 x = a \cos 6x + b \cos 4x + c \cos 2x + d$ finding the constants

b) Hence compute the integral $\int \cos^6 3x \, dx$

Q9. Explain why de Moivre's theorem is useful

Q10. Using de Moivre's theorem prove that

a) $\cos nx = \frac{z^n + z^{-n}}{2}$

b) $\sin nx = \frac{z^n - z^{-n}}{2i}$

Extension

Q11. Find the general expression for $\sin^n x$

Hint: Binomial coefficients are symmetric

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